

Tomasz Kuśmierczyk

Machine Learning · Uncertainty · Bayesian Deep Learning · NLP

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Focus

- Uncertainty & calibration in LLMs
- Bayesian deep learning
- Causal inference
- Reliability & OOD detection

Programming

- **Languages**
Python, C++, SQL, STAN
- **Libraries**
PyTorch, Hugging Face Transformers & PEFT, numpy, pandas, SciPy, GPy
- **Toolkit**
Google Cloud Platform, PySpark, Git

Also

- Open source: code released with every recent paper
github.com/gmum
github.com/tkusmierczyk
- Reviewing for: AAAI 2026 & 2027, UAI 2026, ECML-PKDD 2026, ICML 2024, NeurIPS 2024, AIS-TATS and *Knowledge and Information Systems*
- [Google Scholar](#): 420 citations, h-index 12
- English (fluent), Polish (native), Russian (basic)

Machine learning researcher working on uncertainty, calibration and Bayesian methods for large models. Principal investigator of a national NCN/MSCA project on Bayesian learning for deep neural networks and LLMs, which produced five peer-reviewed papers (+2 under review) and six open-source releases. Earlier work on causal inference from natural experiments at MPI-SWS. PhD from NTNU; publications at NeurIPS, AAAI, EMNLP, UAI, WWW and JMLR. Four years of industry machine learning at Unity. Based in Poland (Kraków/Warsaw).

Experience

Apr 2024 - Mar 2026: Principal Investigator (NCN POLONEZ BIS / MSCA Fellow), Jagiellonian University (Group of Machine Learning Research; project: bayes.ii.uj.edu.pl)

- uncertainty quantification, calibration and out-of-distribution detection for deep networks
- parameter-efficient Bayesian fine-tuning of LLMs: LoRA adapters, projected subspaces, distillation
- set the research agenda and ran the project: budget >1M PLN, recruitment, supervision of a doctoral researcher
- 5 peer-reviewed papers (+2 under review), 6 open-source repositories, public-engagement lectures

Apr 2026 - Aug 2026: Senior Machine Learning Engineer (IC7), Unity Technologies (Helsinki office)

Sep 2020 - Mar 2024: Senior Data Scientist (IC7), Unity Technologies (Helsinki office)

- uncertainty-driven exploration for the conversion prediction model
- in-app purchase model for user life-time value estimation; user-interest models
- large-scale behavioral data: Python, PySpark, Google Cloud Platform

Mar 2018 - Jun 2020: Postdoctoral Researcher, University of Helsinki (Multi-source Probabilistic Inference)

- decision-aware inference: optimizing models against the cost of failures
- prior and hyperparameter elicitation for Bayesian models
- latent categorical variables in variational inference
- co-supervised one master's and one doctoral student; ran a probabilistic-ML reading group

Sep 2016 - Feb 2017: Visiting Researcher, Max Planck Institute for Software Systems (Networks Learning)

- causal effect of platform interventions, identified from natural experiments (WWW 2018, AAAI 2018)

Nov 2013 - Feb 2018: Research Fellow, Norwegian University of Science and Technology (Data and Artificial Intelligence group)

- large-scale time-series mining: correlated bursts across millions of Wikipedia series; user behavior on online platforms

2008 - 2013: Software Engineer / Developer, University of Warsaw (Digital Libraries R&D group; Functional and Structural Genomics) · Diebold Poland

Education

2018: Norwegian University of Science and Technology: PhD in Information Technology

2012: University of Warsaw: MSc in Computer Science

2013 / 2010: University of Warsaw: BSc in Physics · Warsaw Univ. of Technology: BEng in Computer Science

Selected Papers

Bayesian deep learning, calibration and decision-making

- P. Marszałek, **T. Kuśmierczyk***, W. Wydmański, J. Tabor, M. Śmieja*: *ZEUS: Zero-shot Embeddings for Unsupervised Separation of Tabular Data*. **NeurIPS 2025**
Pre-training on a designed prior, so an unseen dataset is handled in a single forward pass.
- M. Sendera*, A. Sorkhei, **T. Kuśmierczyk***: *Revisiting the Equivalence of Bayesian Neural Networks and Gaussian Processes: On the Importance of Learning Activations*. **UAI 2025**
Transfers Gaussian-process priors into Bayesian networks by learning the activation functions.
- E. de Souza da Silva, **T. Kuśmierczyk**, M. Hartmann, A. Klami: *Prior specification via prior predictive matching: Poisson matrix factorization and beyond*. **JMLR 2023**
Chooses priors by matching their observable predictive consequences instead of by posterior inference.
- **T. Kuśmierczyk**, J. Sakaya, A. Klami: *Variational Bayesian Decision-making for Continuous Utilities*. **NeurIPS 2019**
Fits the model against the cost of the decision rather than against the likelihood.
- **T. Kuśmierczyk**, J. Sakaya, A. Klami: *Correcting Predictions for Approximate Bayesian Inference*. **AAAI 2020**
Leaves the fitted posterior alone and corrects the decision rule instead.

Uncertainty in large language models

- P. Marszałek*, K. Bałazy, J. Tabor, **T. Kuśmierczyk***: *Minimal Ranks, Maximum Confidence: Parameter-efficient Uncertainty Quantification for LoRA*. **EMNLP 2025** (Findings)
Halves expected calibration error while training a tenth of LoRA's parameters.
- L. S. H. Nemani, PK Sriji, **T. Kuśmierczyk**: *Efficient Uncertainty in LLMs through Evidential Knowledge Distillation*. **ECML-PKDD 2026**
Distills multi-pass uncertainty-aware teachers into students that need one forward pass.
- V. Dubovik, P. Marszałek, J. Tabor, **T. Kuśmierczyk**: *Bayesian Fine-tuning in Projected Subspaces*. *under review*, **IEEE TPAMI**
Generalizes the above from LoRA to arbitrary projected subspaces of adapter weight space.

Causal inference

- **T. Kuśmierczyk**, M. Gomez-Rodriguez: *On the Causal Effect of Badges*. **WWW 2018**
Causal effect of a platform intervention, identified from a natural experiment. Work done at MPI-SWS.
- **T. Kuśmierczyk**, K. Nøravåg: *On Validation and Predictability of Digital Badges' Influence on Individual Users*. **AAAI 2018**
Whether an intervention's effect on an individual can be validated and predicted from activity traces.

Full list: [Google Scholar](#) · [DBLP](#) · code: github.com/gmum, github.com/tkusmierczyk

Additional Training

Summer schools. **Nordic Probabilistic AI School (ProbAI)**: co-organizer 2019 (Trondheim) and 2022 (Helsinki), head teaching assistant 2022, teaching assistant 2024 and 2026 · DeepLearn 2019, Warsaw · Machine Learning Summer School 2017, Tübingen · RegML 2017, Oslo · Lisbon Machine Learning School 2016 · Gaussian Process Summer School 2015, Sheffield · ECML-PKDD Summer School 2015, Porto

Courses and exchanges. Polish Academy of Sciences, machine learning courses, 2012-2013 · Paris Institute of Technology, Monte Carlo methods (ATHENS program), 2010 · University of Southampton, Erasmus exchange, 2009

Community. ML in PL 2025, the Polish machine learning community conference · MSCA Conference 2024, Mons: Marie Skłodowska-Curie networking conference, attended as a POLONEZ BIS fellow